ETSI TS 118 106 V5.0.1 (2026-08)



TECHNICAL SPECIFICATION

oneM2M;
Management Enablement (BBF)
(oneM2M TS-0006 version 5.0.1 Release 5)

Reference
RTS/oneM2M-000006v5

Keywords
IoT, M2M, management

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from the
ETSI Search & Browse Standards application.

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format on ETSI deliver repository.

Users should be aware that the present document may be revised or have its status changed, this information is available in the Milestones listing.

If you find errors in the present document, please send your comments to the relevant service listed under Committee Support Staff.

If you find a security vulnerability in the present document, please report it through our Coordinated Vulnerability Disclosure (CVD) program.

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part of this document may be reproduced in any form, by any means and in any media, without the prior written authorization of ETSI and except as expressly permitted below.
By way of exception and when the document is a normative deliverable (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)), ETSI authorizes to reproduce and incorporate into products, services and technical documentation only those extracts (e.g. templates) that are strictly necessary for the technical implementation of the normative deliverable, to ensure compliance with the latter.
Nothing in this notice shall be construed as limiting any mandatory exceptions to copyright provided by applicable law.

© ETSI 2026.
All rights reserved.

Contents

Intellectual Property Rights 5
Foreword 5
Modal verbs terminology 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 6
3 Definition of terms, symbols and abbreviations 6
3.1 Terms 6
3.2 Symbols 7
3.3 Abbreviations 7
4 Conventions 7
5 Mapping of basic data types 7
6 Mapping of identifiers 8
6.0 Introduction 8
6.1 Mapping of Device identifiers to the Node Resource 8
6.2 Identifier of an object instance 9
7 Mapping of resources 9
7.0 Introduction 9
7.1 General mapping assumptions 9
7.1.0 Introduction 9
7.1.1 Mapping of Device Identifiers 9
7.1.2 Mapping of Embedded Devices 10
7.2 Resource [deviceInfo] 10
7.3 Resource [memory] 10
7.4 Resource [battery] 11
7.5 Resource [areaNwkInfo] 11
7.6 Resource [areaNwkDeviceInfo] 11
7.7 Resource [eventLog] 12
7.8 Resource [deviceCapability] 12
7.9 Resource [firmware] 13
7.10 Resource [software] 14
7.11 Resource [reboot] 15
7.12 Resource [cmdhPolicy] 15
7.12.0 Introduction 15
7.12.1 Resource [activeCmdhPolicy] 16
7.12.2 Resource [cmdhDefaults] 16
7.12.3 Resource [cmdhDefEcValue] 16
7.12.4 Resource [cmdhEcDefParamValues] 17
7.12.5 Resource [cmdhLimits] 17
7.12.6 Resource [cmdhNetworkAccessRules] 18
7.12.7 Resource [cmdhNwAccessRule] 18
7.12.8 Resource [cmdhBuffer] 19
7.13 Resource Type <mgmtCmd> 19
7.14 Resource Type <execInstance> 20
7.15 Resource [registration] 20
7.16 Resource [dataCollection] 20
7.17 Security Solutions 21
7.17.1 Introduction 21
7.17.2 Resource [authenticationProfile] 21
7.17.3 Resource [trustAnchorCred] 22
7.17.4 Resource [myCertFileCred] 22

7.17.5 Resource [MAFClientRegCfg] 22
7.17.6 Resource [MEFClientRegCfg] 23

8 Mapping of procedures for management 24
8.0 Introduction 24
8.1 Resource Type <mgmtObj> primitive mappings 24
8.1.0 Introduction 24
8.1.1 Alias-Based Addressing Mechanism 24
8.1.2 Create primitive mapping 24
8.1.2.0 Introduction 24
8.1.2.1 M2M Service Layer Resource Instance Identifier mapping 24
8.1.3 Delete primitive mapping 24
8.1.3.1 Delete primitive mapping for deletion of Object Instances 24
8.1.3.2 Delete primitive mapping for software un-install operation 25
8.1.4 Update primitive mapping 26
8.1.4.1 Update primitive mapping for Parameter modifications 26
8.1.4.2 Update primitive mapping for upload file transfer operations 26
8.1.4.3 Update primitive mapping for download file transfer operations 27
8.1.4.4 Update primitive mapping for reboot operation 29
8.1.4.5 Update primitive mapping for factory reset operation 29
8.1.4.6 Update primitive mapping for software install operation 29
8.1.5 Retrieve primitive mapping 30
8.1.6 Notify primitive mapping 31
8.1.6.0 Introduction 31
8.1.6.1 Procedure for subscribed Resource attributes. 31
8.1.6.2 Notification primitive mapping 31
8.2 <mgmtCmd> and <execInstance> resource primitive mappings 32
8.2.1 Update (Execute) primitive for the <mgmtCmd> resource 32
8.2.1.0 Introduction 32
8.2.1.1 Execute File Download 32
8.2.1.2 Execute File Upload Operations 33
8.2.1.3 Report Results using TransferComplete RPC 33
8.2.1.4 Execute Software Operations with ChangeDUState RPC 34
8.2.1.5 Report Results with ChangeDUStateComplete RPC 35
8.2.1.6 Execute Reboot operation 36
8.2.1.7 Execute Factory Reset operation 36
8.2.2 Delete <mgmtCmd> resource primitive mapping 37
8.2.3 Update (Cancel) <execInstance> primitive mapping 37
8.2.4 Delete <execInstance> primitive mapping 38
8.3 Resource [myCertFileCred] primitive mappings 38
8.3.1 Introduction 38
8.3.2 Creation of Resource [myCertFileCred] 38
8.3.2.1 Introduction 38
8.3.2.2 Procedure for creation of Resource [myCertFileCred] 39

9 Server Interactions 40
9.0 Introduction 40
9.1 Communication Session Establishment 40
9.1.1 IN-CSE to ACS Communication Session Establishment 40
9.1.2 ACS to IN-CSE Communication Session Establishment 40
9.1.3 ACS and IN-CSE Communication Session Requirements 40
9.2 Processing of Requests and Responses 41
9.2.1 Request and Notification Formatting 41
9.2.2 ACS Request Processing Requirements 41
9.2.3 ACS Notification Processing Requirements 41
9.3 Discovery and Synchronization of Resources 41
9.4 Access Management 42
9.4.0 Introduction 42
9.4.1 Access Management Requirements 42

10 New Management Technology Specific Resources 42

History 43

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables (European Standard (EN), Technical Specification (TS), Group Specification (GS) or ETSI Standard (ES)) may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI IPR online database.

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™**, **LTE™** and **5G™** logo are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Partnership Project oneM2M (oneM2M).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document describes the protocol mappings between the management Resources for oneM2M and the BBF TR-181 Data Model [6].

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 118 101: "oneM2M; Functional Architecture (oneM2M TS-0001)".

[2] ETSI TS 118 104: "oneM2M; Service Layer Core Protocol Specification (oneM2M TS-0004)".

[3] Void.

[4] BBF TR-069: "CPE WAN Management Protocol", Issue: 1 Amendment 5, November 2013.

[5] BBF TR-106: "Data Model Template for TR-069-Enabled Devices", Issue 1, Amendment 7, September 2013.

[6] BBF TR-181: "Device Data Model for TR-069, Issue 2 Amendment 11", July 2016.

[7] BBF TR-131: "ACS Northbound Interface Requirements, Issue:1", November 2009.

[8] ETSI TS 118 122: "oneM2M; Field Device Configuration (oneM2M TS-0022)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] oneM2M Drafting Rules.

[i.2] ETSI TS 118 111: "oneM2M; Common Terminology (oneM2M TS-0011)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TS 118 111 [i.2] and the following apply:

CPE Proxier: CPE that is capable of proxying the communication between an ACS and a Proxied Device

NOTE: As defined in BBF TR-069 [4].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TS 118 111 [i.2] and the following apply:

ACS	Auto-Configuration Server
ADN	Application Dedicated Node
AE	Application Entity
ASN	Application Service Node
BBF	BroadBand Forum
CMDH	Communication Management and Delivery Handling
CPE	Customer Premise Equipment
CSE	Common Services Entity
CWMP	CPE WAN Management Protocol
DM	Device Management
DU	Deployment Unit
IN-CSE	CSE which resides in the Infrastructure Node
LAN	Local Area Network
MAF	M2M Authentication Function
MN	Middle Node
OUI	Organizationally Unique Identifier
PC	Product Class
RPC	Remote Procedure Call
SN	Serial Number
UPA	Universal Powerline Association
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
USB	Universal Serial Bus
UUID	Universal Unique IDentifier
XML	Extensible Markup Language

4 Conventions

The key words "Shall", "Shall not", "May", "Need not", "Should", "Should not" in the present document are to be interpreted as described in the oneM2M Drafting Rules [i.1].

5 Mapping of basic data types

BBF TR-106 [5] specifies the object structure supported by BBF TR-069 [4] enabled devices and specifies the structural requirements for the data hierarchy. This clause includes the mapping attribute data types to BBF TR-181 [6] parameters which follows the conventions of clause 3 of BBF TR-106 [5] and data types described in Table 4 of BBF TR-106 [5].

Table 5-1: Data Type Mapping

oneM2M Data Types	Mapping to data types in BBF TR-106 [5]	Conversion Notes
xs:boolean	boolean	
xs:string	string	Mapping is constrained to the size of the string.
xs:unsignedInt	unsignedInt	
xs:unsignedLong	unsignedLong	
xs:integer	long	Mapping is constrained to the size of the long data type.
xs:positiveInteger	unsignedLong	Mapping is constrained to a lower limit of 1 and the size of the unsignedLong data type.
xs:nonNegativeInteger	unsignedLong	Mapping is constrained the size of the unsignedLong data type.
Comma separated Lists	Comma separated Lists	Data structure is represented by comma separated list as described in clause 3.2.3 of BBF TR-106 [5].

In some instances the conversion of the contents between data types will cause an error to occur (e.g. xs:integer to long). When an error occurs in the conversion of a data type, the 4000 (BAD_REQUEST) response status code shall be given.

6 Mapping of identifiers

6.0 Introduction

The BBF TR-069 [4] specification defines three (3) types of devices, known as CPEs, that are capable of being managed from the perspective of the BBF TR-069 agent:

- CPE that hosts the BBF TR-069 agent: Table 39 of BBF TR-069 [4] defines the fields required for a CPE to be identified. These fields include the OUI and Serial Number of the CPE assigned by the CPE manufacturer. Optionally the manufacturer may assign a Product Class to the CPE. The format of the identifier is as follows: OUI-[PC-]SN.
- Virtual Device: This type of device is addressed as a CPE. The Virtual Device has its own OUI-[PC-]SN as represented by the CPE Proxier. The CPE Proxier emulates a CWMP agent for each Virtual Device.
- Embedded Device: This type of device is addressed as one or more objects within the data model of the CPE that hosts the BBF TR-069 agent.

6.1 Mapping of Device identifiers to the Node Resource

Node Resources are identified for each instance of an ADN, ASN and MN node and are identified using the M2M Node Identifier (M2M-Node-ID) defined in the oneM2M Functional Architecture [1].

CPE Device identifiers shall map to the nodeID attribute of the <*node*> resource. The CPE Device identifiers are obtained from the contents of the following attributes:

- Device.DeviceInfo.ManufacturerOUI
- Device.DeviceInfo.ProductClass
- Device.DeviceInfo.SerialNumber

Virtual Device identifiers shall map to the nodeID attribute of the <*node*> resource. The Virtual Device identifiers are obtained from the CPE Proxier using the contents of the attributes:

- Device.ManagementServer.VirtualDevice.{i}.ManufacturerOUI
- Device.ManagementServer.VirtualDevice.{i}.ProductClass
- Device.ManagementServer.VirtualDevice.{i}.SerialNumber

Embedded Device identifiers shall map to the nodeID attribute of the <node> resource. The Embedded Device identifiers are obtained using the containing CPE Device or Virtual Device identifiers along with the contents of the attributes of the:

- Device.ManagementServer.EmbeddedDevice.{i}.ControllerID
- Device.ManagementServer.EmbeddedDevice.{i}.ProxiedDeviceID

6.2 Identifier of an object instance

The BBF TR-069 [4] specification permits objects to have multiple object instances where each object instance is contained within the objectPath attribute of the Resource within the context of the Resource's objectId as defined in clause 7.1.

In order to allow the AE or CSE that originated the request that manipulates a Resource to easily align the M2M Service Layer with the Resource's external technology identifier, the value of the object instance "{i}" should be a part of the identifier of the Resource in the M2M Service Layer where possible. For example if the [areaNetwork] resource has an object instance identifier of "Device.X_oneM2M_org_CSE.1.M2MareaNetworkDevice.[foo]" then the M2M Service Layer Resource should be identified using the object instance of the underlying technology (e.g. "/foo" for the Resource areaNetwork).

7 Mapping of resources

7.0 Introduction

This clause contains all information on how to map management resources from ETSI TS 118 104 [2] to managed objects and parameters as defined in the BBF TR-181 [6] data model or the Remote Procedure Calls (RPCs) in BBF TR-069 [4].

7.1 General mapping assumptions

7.1.0 Introduction

BBF TR-069 [4] specifies a protocol for communication between a CPE (Customer Premises Equipment) and an Auto-Configuration Server (ACS). Any BBF TR-069 enabled device has to follow the data model as described in the BBF TR-106 [5] and BBF TR-181 [6] as well as RPCs described in BBF TR-069 [4].

As BBF TR-181 [6] is the model that the Resources are mapped, all Resources shall have the objects of the BBF TR-181 [6] namespace (e.g. "urn:broadband-forum-org:tr-181-2-7-0").

7.1.1 Mapping of Device Identifiers

The Device identifiers for CPEs are mapped to the Resource Types [deviceInfo].

For CPE and Virtual Devices map their Device Identifiers (OUI-[PC-]SN) to the manufacturer, deviceType and deviceLabel attributes of the Resource [deviceInfo].

For Embedded Devices, the ControllerID and ProxiedDeviceID parameters of the Device.ManagementServer.EmbeddedDevice.{i} object instance are mapped to the deviceLabel attribute of the Resource [deviceInfo] as a comma separated list: "Device.ManagementServer.EmbeddedDevice.{i}.ControllerID, Device.ManagementServer.EmbeddedDevice.{i}.ProxiedDeviceID".

7.1.2 Mapping of Embedded Devices

The BBF TR-181 [6] specification does not provide a mechanism where Embedded Devices provide information related to the Device.DeviceInfo objects and sub-objects. Instead the BBF TR-181 [6] provides this information in a manner that is reliant on the Embedded Device's underlying technology (e.g. ZigBee®, UPnP™).

As such the mapping of the [memory] and [battery] Resources are implementation specific for each underlying technology and is outside the scope of the present document.

7.2 Resource [deviceInfo]

The Resource [deviceInfo] is a read-only Resource that shall map to the Device.DeviceInfo object of BBF TR-181 [6] for CPE and Virtual Devices.

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

NOTE: The SerialNumber, ModelNumber, ProductClass attributes for a Virtual device are the same values as the Device.ManagementServer.VirtualDevice.{i} object in the CPE Proxier.

Table 7.2-1: Resource [deviceInfo] for CPE and Virtual Devices

Attribute Name of [deviceInfo]	BBF TR-181 [6] Parameter
deviceLabel	Device.DeviceInfo.SerialNumber
manufacturer	Device.DeviceInfo.Manufacturer
model	Device.DeviceInfo.ModelNumber
deviceType	Device.DeviceInfo.ProductClass
fwVersion	Device.DeviceInfo.SoftwareVersion if the device supports only 1 software version. If the device supports multiple software versions this shall map to Device.DeviceInfo.AdditionalSoftwareVersion
swVersion	Device.DeviceInfo.SoftwareVersion
hwVersion	Device.DeviceInfo.HardwareVersion

Table 7.2-2: Resource [deviceInfo] for Embedded Devices

Attribute Name of [deviceInfo]	BBF TR-181 [6] Parameter
deviceLabel	Comma separated list: Device.ManagementServer.EmbeddedDevice.{i}.ControllerID, Device.ManagementServer.EmbeddedDevice.{i}.ProxiedDeviceID
manufacturer	No mapping available
model	No mapping available
deviceType	No mapping available
fwVersion	No mapping available
swVersion	No mapping available
hwVersion	No mapping available

7.3 Resource [memory]

The Resource [memory] is a read-only Resource that shall map to the Device.DeviceInfo.MemoryStatus object of BBF TR-181 [6] for CPE and Virtual Devices.

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

Attempts to modify the attributes of the memory Resource causes an error code "operation unsupported" to be returned.

Table 7.3-1: Resource [memory]

Attribute Name of [memory]	BBF TR-181 [6] Parameter
memAvailable	Device.DeviceInfo.MemoryStatus.Free
memTotal	Device.DeviceInfo.MemoryStatus.Total

7.4 Resource [battery]

The Resource [battery] is a read-only Resource that shall map to an instance of Device.DeviceInfo.X_oneM2M_org_BatteryStatus.Battery.{i} object for CPE and Virtual Devices.

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

Table 7.4-1: Resource [battery]

Attribute Name of [battery]	BBF TR-181 [6] Parameter
batteryLevel	Device.DeviceInfo.X_oneM2M_org_BatteryStatus.Battery.{i}.Level
batteryStatus	Device.DeviceInfo.X_oneM2M_org_BatteryStatus.Battery.{i}.Status

7.5 Resource [areaNwkInfo]

The Resource [areaNwkInfo] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.M2MareaNetwork.{i} object.

As the Resource [areaNwkInfo] is a multi-instance Resource, the M2MareaNetwork object is a multi-object instance that can be created and deleted.

The M2MareaNetwork instance shall be created using the Add Object RPC of BBF TR-069 [4].

The M2MareaNetwork instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of an M2MareaNetwork shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of an M2MareaNetwork shall be modified using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.5-1: Resource [areaNwkInfo]

Attribute Name of [areaNwkInfo]	X_oneM2M_org Parameter
areaNwkType	Device.X_oneM2M_org_CSE.{i}.M2MareaNetwork.{i}.Type
listOfDevices	Device.X_oneM2M_org_CSE.{i}.M2MareaNetwork.{i}.ListOfDevices

7.6 Resource [areaNwkDeviceInfo]

The Resource [areaNwkDeviceInfo] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.AreaNetworkDevice.{i} object.

As the Resource [areaNwkDeviceInfo] is a multi-instance Resource, the AreaNetworkDevice object is a multi-object instance that can be created and deleted.

Instances of the Resource [areaNwkDeviceInfo] are referenced in the listOfDevices attribute of the associated Resource [areaNwkInfo].

The M2MareaNetworkDevice instance shall be created using the Add Object RPC of BBF TR-069 [4].

The M2MareaNetworkDevice instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of an M2MareaNetworkDevice shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of an M2MareaNetworkDevice shall be modified using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.6-1: Resource [areaNwkDeviceInfo]

Attribute Name of [areaNwkDeviceInfo]	X_oneM2M_org Parameter
devId	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.Host
devType	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.Type
areaNwkId	Reference to Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.M2MareaNetwork
sleepInterval	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.SleepInterval
sleepDuration	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.SleepDuration
status	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.Status
listOfNeighbors	Device.X_oneM2M_org_CSE.{i}.M2MareaNetworkDevice.{i}.Neighbors

7.7 Resource [eventLog]

The Resource [eventLog] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i} object.

The EventLog instance shall be created using the Add Object RPC of BBF TR-069 [4].

The EventLog instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of an EventLog instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of an EventLog instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.7-1: Resource [eventLog]

Attribute Name of [eventLog]	BBF TR-181 [6] Parameter
logTypeId	Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i}.Type
logData	Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i}.Data
logStatus	Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i}.Status
logStart	Set to "True", the Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i}.Enable parameter is set to "True".
logStop	Set to "True", the Device.DeviceInfo.X_oneM2M_org_Diagnostics.EventLog.{i}.Enable parameter is set to "False".

7.8 Resource [deviceCapability]

The Resource [deviceCapability] represents a capability of device that can be administratively enabled or disabled. The lists of capabilities that are managed are defined in the enumeration of the capabilityName attribute. BBF TR-181 [6] data model defines a subset of capabilities listed in the deviceCapability enumeration. The supported device capabilities within BBF TR-181 [6] include:

- LAN Interfaces: USB, Wi-Fi®, HomePlug®, MoCA®, UPA
- Hardware Capabilities: SmartCardReader

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The capabilities shall be enabled and disabled using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.8-1: Resource [deviceCapability]

Attribute Name of [deviceCapability]	BBF TR-181 [6] Parameter
capabilityName	This attribute is fixed based on the value of the capabilityName attribute.
attached	Returns "True"
capabilityActionStatus	Status is defined as: • Success if the SetParameterValues RPC indicates that the operation was successful. • Failure if the response to the SetParameterValues RPCs indicates that the operation failed. • In process if the SetParameterValues RPC is initiated but the response to the SetParameterValues RPC has not been received.
currentState	USB: Device.USB.Interface.{i}.Enable Wi-Fi: Device.Wi-Fi.Radio.{i}.Enable HomePlug: Device.HomePlug.Interface.{i}.Enable MoCA: Device.MoCA.Interface.{i}.Enable UPA: Device.UPA.Interface.{i}.Enable SmartCardReader: Device.SmartCardReaders.SmartCardReader.{i}.Enable
enable	USB: Device.USB.Interface.{i}.Enable Wi-Fi: Device.Wi-Fi.Radio.{i}.Enable HomePlug: Device.HomePlug.Interface.{i}.Enable MoCA: Device.MoCA.Interface.{i}.Enable UPA: Device.UPA.Interface.{i}.Enable SmartCardReader: Device.SmartCardReaders.SmartCardReader.{i}.Enable
disable	Same parameter is used to disable a capability as the enable attribute.

7.9 Resource [firmware]

The Resource [firmware] represents a firmware instance and is not considered a BBF TR-069 [4] managed entity within the device until the firmware Resource's update attribute has been written a value of "True". When this occurs, the BBF TR-069 [4] Download RPC shall be invoked.

NOTE: In many instances, the server from which the firmware is downloaded requires authentication in the form of Username and Password credentials. The CSE that executes firmware download shall maintain the mapping of the username and password of the download server needed to download the firmware outside the lifecycle of the specific firmware.

Table 7.9-1: Resource [firmware]

Attribute Name of [firmware]	RPC Download Arguments
URL	URL
update	When set to the value of "True" executes the Download operations with a FileType "1 Firmware Upgrade Image".
	Username: Received from the CSE for the download server where the update is set to "True".
	Password: Received from the CSE for the download server where the update is set to "True".
	CommandKey: Automatically set by the CSE where the update is set to "True" in order to correlate the TransferComplete response.
	FileSize: 0 (not used)
	TargetFileName: <empty> (not used)
	DelaySeconds: 0 (immediate)
	SuccessURL: <empty> (not used)
	FailureURL: <empty> (not used)

7.10 Resource [software]

The Resource [software] is a multi-instance Resource where each instance of the Resource maps directly to an instance of Device.SoftwareModules.DeploymentUnit.{i} object for the deployment aspects (install, uninstall) of the Resource [software]. The install and uninstall operation of the Resource [software] is performed using a combination of the ChangeDUState and ChangeDUStateComplete RPCs.

Once a Resource [software] has been installed, the Resource shall be mapped to the associated Device.SoftwareModules.ExecutionUnit.{i} objects in order to activate and deactivate the associated execution unit.

The Resource [software] version and name shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The activate and deactivate operations of the Resource [software] shall be performed by manipulating the Device.SoftwareModules.ExecutionUnit.{i}.RequestedState parameter using the SetParameterValues RPC.

NOTE: The Resource [software] provides support for only 1 Execution Unit per Deployment Unit. If a Deployment Unit is discovered by the M2M Service Layer that contains multiple Execution Units for a Deployment Unit; only 1 Execution Unit is exposed. The selection of which Execution Unit is implementation specific.

Table 7.10-1: Resource [software]

Attribute Name of [software]	Description
version	Device.SoftwareModules.DeploymentUnit.{i}Version
name	Device.SoftwareModules.DeploymentUnit.{i}.Name
URL	Device.SoftwareModules.DeploymentUnit.{i}.URL
install	Use the ChangeDUState:InstallOpStruct
installStatus	Status is defined as: • Success if the ChangeDUStateComplete RPC indicates that the operation was successful. • Failure if the response to the ChangeDUState or ChangeDUStateComplete RPCs indicates that the operation failed. • In process if the ChangeDUState RPC is initiated but the ChangeDUStateComplete RPC has not been received.
activate	The action that activates software previously installed.
deactivate	The action that deactivates software.
activeStatus	Status is defined as: • Success if the SetParameterValues RPC indicates that the operation was successful. • Failure if the response to the SetParameterValues RPCs indicates that the operation failed. • In process if the SetParameterValues RPC is initiated but the response to the SetParameterValues RPC has not been received.

Table 7.10-2: RPC ChangeDUState:InstallOpStruct Arguments

Name	Notes
URL	URL of the Server that M2M Node uses to download the DU.
Username	Username credential of Server that the CPE uses to download the DU - Supplied by the CSE.
Password	Password credential of Server that the CPE uses to download the DU - Supplied by the CSE.
UUID	Supplied by the CSE and used to correlate the DU for the uninstall operation.
ExecutionEnvRef	<empty> not used

Table 7.10-3: RPC ChangeDUState:UninstallOpStruct Arguments

Name	Notes
UUID	UUID of the DU that was installed - Maintained by the CSE.
ExecutionEnvRef	<empty> not used

7.11 Resource [reboot]

The Resource [reboot] maps to either the Reboot RPC or FactoryReset RPC of BBF TR-069 [4].

When the reboot attribute of the Resource [reboot] is set to "True", the CSE shall execute the Reboot RPC of BBF TR-069 [4].

When the factoryReset attribute of Resource [reboot] is set to "True", the CSE shall execute the FactoryReset RPC of BBF TR-069 [4].

Table 7.11-1: Resource [reboot]

Attribute Name of [reboot]	Description
reboot	Executes the Reboot RPC
factoryReset	FactoryReset RPC

Table 7.11-2: RPC Reboot Arguments

Name	Notes
CommandKey	Automatically set by the CSE where the reboot is set to "True" in order to correlate the "M-Reboot" Event from the next Inform.

7.12 Resource [cmdhPolicy]

7.12.0 Introduction

The Resource [cmdhPolicy] represents a set of rules defining which CMDH parameters will be used by default when a request issued by a local originator contains the **ec** (event category) parameter but not all other CMDH parameters, see clause D.12 of ETSI TS 118 101 [1].

The Resource [cmdhPolicy] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i} object.

The Policy instance shall be created using the Add Object RPC of BBF TR-069 [4].

The Policy instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a Policy instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a Policy instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12-1: Resource [cmdhPolicy]

Attribute Name of [cmdhPolicy]	X_oneM2M_org Parameter
name	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.Name
cmdhDefaults	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.DefaultRule
cmdhLimits	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.LimitRules
cmdhNetworkAccessRules	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.NetworkAccessECRules
cmdhBuffer	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.BufferRules

7.12.1 Resource [activeCmdhPolicy]

The Resource [activeCmdhPolicy] provides a link to the currently active set of CMDH policies, see clause D.12.1 of ETSI TS 118 101 [1].

The Resource [activeCmdhPolicy] is mapped to the Enable parameter of the Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i} object.

The information of a Policy instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.1-1: Resource [activeCmdhPolicy]

Attribute Name of [activeCmdhPolicy]	X_oneM2M_org Parameter
cmdhPolicy	Device.X_oneM2M_org_CSE.{i}.CMDH.Policy.{i}.Enable At most one Policy instance shall be enabled at a time. As such the Policy instance that has the Enable parameter with a value of "True" is the active CMDH policy.

7.12.2 Resource [cmdhDefaults]

The Resource [cmdhDefaults] defines default CMDH policy values, see clause D.12.2 of ETSI TS 118 101 [1].

The Resource [cmdhDefaults] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.Default.{i} object.

The Default instance shall be created using the Add Object RPC of BBF TR-069 [4].

The Default instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a Default instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a Default instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.2-1: Resource [cmdhDefaults]

Attribute Name of [cmdhDefaults]	X_oneM2M_org Parameter
cmdhDefEcValue	Device.X_oneM2M_org_CSE.{i}.CMDH.Default.{i}.DefaultECRules
cmdhEcDefParamValues	Device.X_oneM2M_org_CSE.{i}.CMDH.Default.{i}.DefaultECParamRules

7.12.3 Resource [cmdhDefEcValue]

The Resource [cmdhDefEcValue] represents a value for the **ec** (event category) parameter of an incoming request, see clause D.12.3 of ETSI TS 118 101 [1].

The Resource [cmdhDefEcValue] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i} object.

The DefaultECRule instance shall be created using the Add Object RPC of BBF TR-069 [4].

The DefaultECRule instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a DefaultECRule instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a DefaultECRule instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.3-1: Resource [cmdhDefEcValue]

Attribute Name of [cmdhDefEcValue]	X_oneM2M_org Parameter
order	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.Order
defEcValue	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.EventCategory
requestOrigin	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.RequestOrigin
requestContext	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.RequestContext
requestContextNotification	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.RequestContextNotificationEnable
requestCharacteristics	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECRule.{i}.RequestCharacteristics

7.12.4 Resource [cmdhEcDefParamValues]

The Resource [cmdhEcDefParamValues] represents a specific set of default values for the CMDH related parameters **rqet** (request expiration timestamp), **rset** (result expiration timestamp), **oet** (operational execution time), **rp** (response persistence) and **da** (delivery aggregation) that are applicable for a given **ec** (event category) if these parameters are not specified in the request, see clause D.12.4 of ETSI TS 118 101 [1].

The Resource [cmdhEcDefParamValues] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i} object.

The DefaultECParamRule instance shall be created using the Add Object RPC of BBF TR-069 [4].

The DefaultECParamRule instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a DefaultECParamRule instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a DefaultECParamRule instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.4-1: Resource [cmdhEcDefParamValues]

Attribute Name of [cmdhEcDefParamValues]	X_oneM2M_org Parameter
applicableEventCategory	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.EventCategories
defaultRequestExpTime	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.RequestExpTime
defaultResultExpTime	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.ResultExpTime
defaultOpExecTime	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.OperationExecTime
defaultRespPersistence	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.ResponsePersistence
defaultDelAggregation	Device.X_oneM2M_org_CSE.{i}.CMDH.DefaultECParamRule.{i}.DeliveryAggregation

7.12.5 Resource [cmdhLimits]

The Resource [cmdhLimits] represents limits for CMDH related parameter values, see clause D.12.5 of ETSI TS 118 101 [1].

The Resource [cmdhLimits] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i} object.

The Limit instance shall be created using the Add Object RPC of BBF TR-069 [4].

The Limit instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a Limit instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a Limit instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.5-1: Resource [cmdhLimits]

Attribute Name of [cmdhLimits]	X_oneM2M_org Parameter
order	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.Order
requestOrigin	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.RequestOrigin
requestContext	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.RequestContext
requestContextNotification	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.RequestContextNotificationEnable
requestCharacteristics	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.RequestCharacteristics
limitsEventCategory	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.EventCategories
limitsRequestExpTime	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.RequestExpTime
limitsResultExpTime	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.ResultExpTime
limitsOpExecTime	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.OperationExecTime
limitsRespPersistence	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.ResponsePersistence
limitsDelAggregation	Device.X_oneM2M_org_CSE.{i}.CMDH.Limit.{i}.DeliveryAggregation

7.12.6 Resource [cmdhNetworkAccessRules]

The Resource [cmdhNetworkAccessRules] defines the usage of underlying networks for forwarding information to other CSEs during processing of CMDH-related requests in a CSE, see clause D.12.6 of ETSI TS 118 101 [1].

The Resource [cmdhNetworkAccessRules] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessECRule.{i} object.

The NetworkAccessECRule instance shall be created using the Add Object RPC of BBF TR-069 [4].

The NetworkAccessECRule instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a NetworkAccessECRule instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a NetworkAccessECRule instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.6-1: Resource [cmdhNetworkAccessRules]

Attribute Name of [cmdhNetworkAccessRules]	X_oneM2M_org Parameter
applicableEventCategories	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessECRule.{i}.EventCategories
cmdhNwAccessRule	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessECRule.{i}.NetworkAccessRules

7.12.7 Resource [cmdhNwAccessRule]

The Resource [cmdhNwAccessRule] define limits in usage of specific underlying networks for forwarding information to other CSEs during processing of CMDH-related requests, see clause D.12.7 of ETSI TS 118 101 [1].

The Resource [cmdhNwAccessRule] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i} object.

The NetworkAccessRule instance shall be created using the Add Object RPC of BBF TR-069 [4].

The NetworkAccessRule instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a NetworkAccessRule instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a NetworkAccessRule instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.7-1: Resource [cmdhNwAccessRule]

Attribute Name of [cmdhNwAccessRule]	X_oneM2M_org Parameter
targetNetwork	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.TargetNetworks
minReqVolume	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.MinimumReqVolume
backOffParameters	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.BackoffTime
	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.BackoffTimeIncrement
	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.MaximumBackoffTime
otherConditions	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.OtherConditions
allowedSchedule	Device.X_oneM2M_org_CSE.{i}.CMDH.NetworkAccessRule.{i}.AllowedSchedule

7.12.8 Resource [cmdhBuffer]

The Resource [cmdhBuffer] represents limits in usage of buffers for temporarily storing information that needs to be forwarded to other CSEs during processing of CMDH-related requests in a CSE, see clause D.12.8 of ETSI TS 118 101 [1].

The Resource [cmdhBuffer] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_CSE.{i}.CMDH.Buffer.{i} object.

The Buffer instance shall be created using the Add Object RPC of BBF TR-069 [4].

The Buffer instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a Buffer instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a Buffer instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.12.8-1: Resource [cmdhBuffer]

Attribute Name of [cmdhBuffer]	X_oneM2M_org Parameter
applicableEventCategory	Device.X_oneM2M_org_CSE.{i}.CMDH.Buffer.{i}.EventCategories
maxBufferSize	Device.X_oneM2M_org_CSE.{i}.CMDH.Buffer.{i}.MaximumBufferSize
storagePriority	Device.X_oneM2M_org_CSE.{i}.CMDH.Buffer.{i}.StoragePriority

7.13 Resource Type <mgmtCmd>

Each mgmtCmd Resource shall map to BBF TR-069 RPC commands based on the value of cmdType. Accordingly, execReqArgs shall contain arguments related to the corresponding BBF TR-069 RPCs. The details about corresponding procedure mapping are described in clause 8.2.

Table 7.13-1: Resource Type <mgmtCmd>

Attribute cmdType of mgmtCmd	Attribute execReqArgs of mgmtCmd
cmdType = RESET	Shall include all arguments related to BBF FactoryReset RPC
cmdType = REBOOT	Shall include all arguments related to BBF Reboot RPC
cmdType = UPLOAD	Shall include all arguments related to BBF Reboot RPC
cmdType = DOWNLOAD	Shall contain all arguments related to BBF Reboot RPC
cmdType = SOFTWAREINSTALL	Shall contain all arguments related to BBF ChangeDUState RPC which shall contain "InstallOpStruct" structure.
cmdType = SOFTWAREUNINSTALL	Shall contain all arguments related to BBF ChangeDUState RPC which shall contain "UninstallOpStruct" structure.

7.14 Resource Type <execInstance>

The <execInstance> resource from ETSI TS 118 104 [2] shall map to BBF CancelTransfer RPC commands when it is disabled/cancelled using a Update operation or deleted using a Delete operation. The details are described in clause 8.2.

7.15 Resource [registration]

The Resource [registration] represents the configuration information needed to register and AE or CSE with a Registrar CSE.

The Resource [registration] is a multi-instance object where the key of the object is the originatorID (i.e. AE-ID, CSE-ID). The following rules are used to determine the object instance based on the originatorID:

- When the originatorID resource is for a CSE-ID, the BBF TR-069 [4] object instance Device.X_oneM2M_org_CSE.{i} shall be used for the specified CSE-ID.
- When the originatorID resource is for an AE-ID, the BBF TR-069 [4] object instance Device.X_oneM2M_org_AE.{i} shall be used for the specified AE-ID.

The information shall be created using the Add Object RPC of BBF TR-069 [4].

The information shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The capabilities shall be enabled and disabled using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.15-1: Resource [registration] for CSE

Attribute Name of [registration]	BBF TR-181 [6] Parameter (X_oneM2M_org_CSE)
originatorID	ID - See description of the type of object to instantiate.
poA	PointOfAccess
CSEBase	CSEBase - Resource-ID of the CSEBase of Registrar CSE.
CSE-ID	CSEID - CSE-ID of the Registrar CSE.
externalID	ExternalID
triggerRecipientID	TriggerRecipientID
mgmtLink [authenticationProfile]	AuthenticationProfile (BBF TR-069 [4] reference parameter that references a row in the Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile table)

Table 7.15-2: Resource [registration] for AE

Attribute Name of [registration]	BBF TR-181 [6] Parameter (X_oneM2M_org_AE)
originatorID	ID - See description of the type of object to instantiate.
poA	PointOfAccess
CSEBase	CSEBase - Resource-ID of the CSEBase of Registrar CSE.
CSE-ID	CSEID - CSE-ID of the Registrar CSE.
appID	ApplicationID
mgmtLink [authenticationProfile]	AuthenticationProfile (BBF TR-069 [4] reference parameter that references a row in the Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile table)

7.16 Resource [dataCollection]

The Resource [dataCollection] represents data collection (measurement) and transmittal (reporting) properties for an AE.

The information shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The capabilities shall be enabled and disabled using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.16-1: Resource [dataCollection]

Attribute Name of [dataCollection]	BBF TR-181 [6] Parameter
containerPath	ContainerPath
reportingSchedule	ReportingSchedule
measurementSchedule	CollectionSchedule

7.17 Security Solutions

7.17.1 Introduction

This clause contains information on how to map the security specific management resources from ETSI TS 118 122 [8] to managed objects and parameters as defined in the BBF TR-181 [6] data model or the Remote Procedure Calls (RPCs) in BBF TR-069 [4].

7.17.2 Resource [authenticationProfile]

The Resource [authenticationProfile] represents configuration information regarding establishing mutually-authenticated secure communications. The security principal using this configuration information can be a CSE or AE or the Managed ADN/ASN/MN acting as security principal on behalf of AEs on the Node, see clause 7.1.4 of ETSI TS 118 122 [8].

The Resource [authenticationProfile] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile.{i} object.

The AuthenticationRule instance shall be created using the Add Object RPC of BBF TR-069 [4].

The AuthenticationRule instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a AuthenticationProfile instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a AuthenticationProfile instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.17.2-1: Resource [authenticationProfile]

Attribute Name of [authenticationProfile]	Parameters of Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile.{i}
SUID	SUID
TLSCiphersuites	TLSCiphersuites
symmKeyID	SymmetricKeyID
symmKeyValue	SymmetricKeyValue
MAFKeyRegLabels	MAFKeyRegLabels
MAFKeyRegDuration	MAFKeyRegDuration
mycertFingerprint	MyCert (reference)
rawPubKeyID	RawPubKeyID
mgmtLink [trustAnchorCred]	TrustAnchorCredentials (list of references)

The parameter MyCert is a BBF TR-069 [4] reference parameter that references a row in the Device.Security.Certificate table where the value of the mycertFingerprint attribute matches the value of a Device.Security.Certficate.{i}.X_oneM2M_org_Fingerprint parameter. The X_oneM2M_org_Fingerprint parameter shall be a unique key for the Device.Security.Certificate table.

The parameter TrustAnchorCredentials is a list of BBF TR-069 reference parameter where each entry in the list references a row in the Device.X_oneM2M_org_SecuritySolution.TrustAnchorCredential table.

7.17.3 Resource [trustAnchorCred]

The Resource [trustAnchorCred] represents configuration information regarding certificates provided by certificate authorities used be managed entities to authenticate peer endpoints, see clause 7.1.6 of ETSI TS 118 122 [8].

The Resource [trustAnchorCred] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_SecuritySolution.TrustAnchorCredential.{i} object.

The TrustAnchorCredential instance shall be created using the Add Object RPC of BBF TR-069 [4].

The TrustAnchorCredential instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a TrustAnchorCredential instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a TrustAnchorCredential instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.17.3-1: Resource [trustAnchorCred]

Attribute Name of [trustAnchorCred]	Parameters of Device.X_oneM2M_org_SecuritySolution.TrustAnchorCredential.{i}
certFingerprint	Fingerprint
URI	RemoteTrustStore

7.17.4 Resource [myCertFileCred]

The Resource [myCertFileCred] represents configuration information regarding certificates presented by the managed entity to remote entities for the establishment of secure communications, see clause 7.1.5 of ETSI TS 118 122 [8].

The Resource [myCertFileCred] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.Security.Certificate.{i} object.

The Certificate instance shall be created either using the Download RPC of BBF TR-069 [4] or via an out-of-band mechanism.

The Certificate instance shall be deleted using the Download RPC of BBF TR-069 [4] or via an out-of-band mechanism.

The information of a Certificate instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a Certificate instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.17.4-1: Resource [myCertFileCred]

Attribute Name of [myCertFileCred]	Parameters of Device.Security.Credential.{i}
SUIDs	X_oneM2M_org_SUIDs
myCertFileFormat	X_oneM2M_org_Format
myCertFileContent	The certificate is downloaded as part of the Download RPC of BBF TR-069 [4]

The parameter AuthenticationProfile is a BBF TR-069 reference parameter that references a row in the Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile table where the value of the mycertFingerprint attribute matches the value of a Device.Security.Certficate.{i}.X_oneM2M_org_Fingerprint parameter. The X_oneM2M_org_Fingerprint parameter shall be a unique key for the Device.Security.Certificate table.

7.17.5 Resource [MAFClientRegCfg]

The Resource [MAFClientRegCfg] represents configuration information that permits a MAF client to register with a MAF, see clause 7.1.7 of ETSI TS 118 122 [8].

The Resource [MAFClientRegCfg] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_SecuritySolution.MAFClientRegistration.{i} object.

The MAFClientRegistration instance shall be created using the Add Object RPC of BBF TR-069 [4].

The MAFClientRegistration instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a MAFClientRegistration instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a MAFClientRegistration instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.17.5-1: Resource [MAFClientRegCfg]

Attribute Name of [MAFClientRegCfg]	Parameters of Device.X_oneM2M_org_SecuritySolution.MAFClientRegistration.{i}
mgmtLink [authenticationProfile]	AuthenticationProfile (BBF TR-069 [4] reference parameter that references a row in the Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile table)
fqdn	FQDN
adminFQDN	AdminFQDN
httpPort	HTTPPort
coapPort	CoAPPort
websocketPort	WebsocketPort
expirationTime	ExpirationTimeStamp

7.17.6 Resource [MEFClientRegCfg]

The Resource [MEFClientRegCfg] represents configuration information that permits a MEF client to register with a MEF, see clause 7.1.8 of ETSI TS 118 122 [8].

The Resource [MEFClientRegCfg] is a multi-instance Resource where each instance of the Resource shall map to an instance of Device.X_oneM2M_org_SecuritySolution.MEFClientRegistration.{i} object.

The MEFClientRegistration instance shall be created using the Add Object RPC of BBF TR-069 [4].

The MEFClientRegistration instance shall be deleted using the Delete Object RPC of BBF TR-069 [4].

The information of a MEFClientRegistration instance shall be retrieved using the GetParameterValues RPC of BBF TR-069 [4].

The information of a MEFClientRegistration instance shall be updated using the SetParameterValues RPC of BBF TR-069 [4].

Table 7.17.6-1: Resource [MEFClientRegCfg]

Attribute Name of [MEFClientRegCfg]	Parameters of Device.X_oneM2M_org_SecuritySolution.MEFClientRegistration.{i}
mgmtLink [authenticationProfile]	AuthenticationProfile (BBF TR-069 [4] reference parameter that references a row in the Device.X_oneM2M_org_SecuritySolution.AuthenticationProfile table)
fqdn	FQDN
adminFQDN	AdminFQDN
httpPort	HTTPPort
coapPort	CoAPPort
websocketPort	WebsocketPort
expirationTime	ExpirationTimeStamp

8 Mapping of procedures for management

8.0 Introduction

This clause contains all information on how to map management resource primitives from ETSI TS 118 104 [2] to the Remote Procedure Calls (RPCs) in BBF TR-069 [4].

8.1 Resource Type <mgmtObj> primitive mappings

8.1.0 Introduction

This clause contains all information on how to map Resource Type <mgmtObj> primitives from ETSI TS 118 104 [2] to the Remote Procedure Calls (RPCs) in BBF TR-069 [4].

8.1.1 Alias-Based Addressing Mechanism

In order to utilize the Alias-Based Addressing Mechanism, the mechanism has to be supported by the ACS and CPE in order to map the M2M Service Layer identifier for the Resource instance to the CPE object instance. If the AliasBased Addressing Mechanism feature is not supported by either the ACS or CPE, the CSE has to retain the mapping of the these M2M Resource instance identifiers.

8.1.2 Create primitive mapping

8.1.2.0 Introduction

The Create Request and Response primitives shall map to the AddObject RPC. The AddObject RPC is defined in BBF TR-069 [4] as a synchronous RPC and returns a successful response or one of the following fault codes in Table 8.1.2.0-1.

Table 8.1.2.0-1: AddObject Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9005	Invalid Parameter name (associated with Set/GetParameterValues, GetParameterNames, Set/GetParameterAttributes, AddObject, and DeleteObject)	5001 (NOT_IMPLEMENTED)

8.1.2.1 M2M Service Layer Resource Instance Identifier mapping

When the Resource is a multi-instance Resource, the AddObject RPC should utilize the Alias-Based Addressing Mechanism as defined in clause 3.6.1 of BBF TR-069 [4] in order to use the Resource instance value of the URI.

8.1.3 Delete primitive mapping

8.1.3.1 Delete primitive mapping for deletion of Object Instances

The Delete Request and Response primitives that results in the deletion of a Resource shall map to the DeleteObject RPC. The DeleteObject RPC is defined in BBF TR-069 [4] as a synchronous RPC and returns a successful response or one of the following fault codes in Table 8.1.3.1-1.

Table 8.1.3.1-1: DeleteObject Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9005	Invalid Parameter name (associated with Set/GetParameterValues, GetParameterNames, Set/GetParameterAttributes, AddObject, and DeleteObject)	5001 (NOT_IMPLEMENTED)

8.1.3.2 Delete primitive mapping for software un-install operation

The Delete Request and Response primitives that results in a software un-install operation (e.g. Resource [software]) shall use the ChangeDUState mechanism defined in BBF TR-069 [4]. The ChangeDUState mechanism is an asynchronous command that consists of the synchronous ChangeDUState RPC for the un-installation request and the asynchronous ChangeDUStateComplete RPC. The ChangeDUState RPC returns a successful response or one of the following fault codes in Table 8.1.3.2-1. A successful response means that the CPE has accepted the ChangeDUState RPC.

Table 8.1.3.2-1: ChangeDUState Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)

Once the CPE has attempted to change the state of the deployment unit, the CPE reports the result of the state change operation using the ChangeDUStateComplete RPC. The ChangeDUStateComplete RPC indicates a successful operation or one of the following fault codes in Table 8.1.3.2-2.

Table 8.1.3.2-2: ChangeDUStateComplete Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9022	Invalid UUID Format (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	4000 (BAD_REQUEST)
9023	Unknown Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	4000 (BAD_REQUEST)
9024	Disabled Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	4000 (BAD_REQUEST)

Fault code	Description	Response Status Code
9025	Deployment Unit to Execution Environment Mismatch (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	4000 (BAD_REQUEST)
9026	Duplicate Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	4000 (BAD_REQUEST)
9027	System Resources Exceeded (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	4000 (BAD_REQUEST)
9028	Unknown Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update and Uninstall)	4000 (BAD_REQUEST)
9029	Invalid Deployment Unit State (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update and Uninstall)	4000 (BAD_REQUEST)
9030	Invalid Deployment Unit Update - Downgrade not permitted (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)
9031	Invalid Deployment Unit Update - Version not specified (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)
9032	Invalid Deployment Unit Update - Version already exists (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)

8.1.4 Update primitive mapping

8.1.4.1 Update primitive mapping for Parameter modifications

The Update Request and Response primitives that modifies the value of Resource attributes shall map to the SetParameterValues RPC. The SetParametersValue RPC is defined in BBF TR-069 [4] as a synchronous RPC and returns a successful response or one of the following fault codes in Table 8.1.4.1-1.

Table 8.1.4.1-1: SetParameterValues Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9005	Invalid Parameter name (associated with Set/GetParameterValues, GetParameterNames, Set/GetParameterAttributes, AddObject, and DeleteObject)	5001 (NOT_IMPLEMENTED)
9006	Invalid Parameter type (associated with SetParameterValues)	4000 (BAD_REQUEST)
9007	Invalid Parameter value (associated with SetParameterValues)	4000 (BAD_REQUEST)
9008	Attempt to set a non-writable Parameter (associated with SetParameterValues)	4000 (BAD_REQUEST)

8.1.4.2 Update primitive mapping for upload file transfer operations

The Update Request and Response primitives that results in an upload file transfer operation (e.g. logStop attribute of the Resource [eventLog]) shall use the Upload mechanism defined in BBF TR-069 [4]. The Upload mechanism is an asynchronous command that consists of the synchronous Upload RPC for the Upload and the asynchronous TransferComplete RPC. The Upload RPC returns a successful response or one of the following fault codes in Table 8.1.4.2-1. A successful response means that the CPE has accepted the Upload RPC.

Table 8.1.4.2-1: Upload Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)

Once the CPE has attempted to upload the file, the CPE reports the result of the Upload operation using the TransferComplete RPC. The TransferComplete RPC indicates a successful operation or one of the following fault codes in Table 8.1.4.2-2.

Table 8.1.4.2-2: TransferComplete Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9014	File transfer failure: unable to join multicast group (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9019	File transfer failure: file authentication failure (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9020	File transfer failure: unable to complete download within specified time windows (associated with TransferComplete method).	4000 (BAD_REQUEST)

8.1.4.3 Update primitive mapping for download file transfer operations

The Update Request and Response primitives that results in a download file transfer operation (e.g. update attribute of Resource [firmware]) shall use the Download mechanism defined in BBF TR-069 [4]. The Download mechanism is an asynchronous command that consists of the synchronous Download RPC for the Download and the asynchronous

TransferComplete RPC. The Download RPC returns a successful response or one of the following fault codes in Table 8.1.4.3-1. A successful response means that the CPE has accepted the Download RPC.

Table 8.1.4.3-1: Download Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)

Once the CPE has attempted to download the file, the CPE reports the result of the download operation using the TransferComplete RPC. The TransferComplete RPC indicates a successful operation or one of the following fault codes in Table 8.1.4.3-2.

Table 8.1.4.3-2: TransferComplete Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9014	File transfer failure: unable to join multicast group (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9019	File transfer failure: file authentication failure (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9020	File transfer failure: unable to complete download within specified time windows (associated with TransferComplete method).	4000 (BAD_REQUEST)

8.1.4.4 Update primitive mapping for reboot operation

The Update Request and Response primitives that results in a reboot operation (e.g. reboot attribute of Resource [reboot]) shall use the Reboot RPC defined in BBF TR-069 [4]. The Reboot RPC is asynchronous command. The Reboot RPC returns a successful response or one of the following fault codes in Table 8.1.4.4-1.

Table 8.1.4.4-1: Reboot Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)

8.1.4.5 Update primitive mapping for factory reset operation

The Update Request and Response primitives that results in a factory reset operation (e.g. factoryReset attribute of Resource [reboot]) shall use the FactoryReset RPC defined in BBF TR-069 [4]. The FactoryReset RPC is an asynchronous command. The FactoryReset RPC returns a successful response or one of the following fault codes in Table 8.1.4.5-1.

Table 8.1.4.5-1: FactoryReset Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)

8.1.4.6 Update primitive mapping for software install operation

The Update Request and Response primitives that results in a software installation operation (e.g. install attribute of Resource [software]) shall use the ChangeDUState mechanism defined in BBF TR-069 [4]. The ChangeDUState mechanism is an asynchronous command that consists of the synchronous ChangeDUState RPC for the download and the asynchronous ChangeDUStateComplete RPC. The ChangeDUState RPC returns a successful response or one of the following fault codes in Table 8.1.4.6-1. A successful response means that the CPE has accepted the ChangeDUState RPC.

Table 8.1.4.6-1: ChangeDUState Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)

Once the CPE has attempted to change the state of the deployment unit, the CPE reports the result of the state change operation using the ChangeDUStateComplete RPC. The ChangeDUStateComplete RPC indicates a successful operation or one of the following fault codes in Table 8.1.4.6-2.

Table 8.1.4.6-2: ChangeDUStateComplete Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)

Fault code	Description	Response Status Code
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9022	Invalid UUID Format (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	4000 (BAD_REQUEST)
9023	Unknown Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	4000 (BAD_REQUEST)
9024	Disabled Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	4000 (BAD_REQUEST)
9025	Deployment Unit to Execution Environment Mismatch (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	4000 (BAD_REQUEST)
9026	Duplicate Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	4000 (BAD_REQUEST)
9027	System Resources Exceeded (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	4000 (BAD_REQUEST)
9028	Unknown Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update and Uninstall)	4000 (BAD_REQUEST)
9029	Invalid Deployment Unit State (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update and Uninstall)	4000 (BAD_REQUEST)
9030	Invalid Deployment Unit Update - Downgrade not permitted (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)
9031	Invalid Deployment Unit Update - Version not specified (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)
9032	Invalid Deployment Unit Update - Version already exists (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	4000 (BAD_REQUEST)

8.1.5 Retrieve primitive mapping

The Retrieve Request and Response primitives shall map to the GetParameterValues RPC. The GetParametersValue RPC is defined in BBF TR-069 [4] as a synchronous RPC and returns a successful response or one of the following fault codes in Table 8.1.5-1.

Table 8.1.5-1: GetParameterValues Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot used to indicate Parameters in error)	4000 (BAD_REQUEST)
9005	Invalid Parameter name (associated with Set/GetParameterValues, GetParameterNames, Set/GetParameterAttributes, AddObject, and DeleteObject)	4000 (BAD_REQUEST)

8.1.6 Notify primitive mapping

8.1.6.0 Introduction

The NotifyRequest and Response primitives permit notifications to AE or CSEs that have subscribed to a Resource.

While BBF TR-069 [4] has the capability to notify the subscribed ACS when an object's parameter has been modified, BBF TR-069 [4] does not have the capability for an ACS to be notified if any parameter within the object has been modified unless the ACS individually subscribes to all the parameters of the object.

As such the procedure for mapping the Notify Request and Response primitives for BBF TR-069 [4] is not possible unless the CSE subscribes to receive notification to all the parameters of an Object that are mapped to the Resource's attributes.

NOTE: In many implementations, subscribing to all the parameters of an Object that are mapped to the Resource can cause performance issues in the CPE as well as the CSE. As such using the attribute based subscription capabilities of BBF TR-069 [4] for subscription of Resources should be avoided when possible.

8.1.6.1 Procedure for subscribed Resource attributes.

When a <subscription> Resource for a <mgmtObj> Resource is Created, Deleted or Updated the CSE shall map to the SetParameterAttributes RPC in the following manner:

- BBF TR-069 [4] provides the capability to subscribe to changes of a specific attribute through the use of the SetParameterAttributes RPC using the "Active" value for the Notification parameter.
- BBF TR-069 [4] provides the capability to un-subscribe to changes of a specific attribute through the use of the SetParameterAttributes RPC using the "None" value for the Notification parameter.

The SetParametersAttributes RPC is defined in BBF TR-069 [4] as a synchronous RPC and returns a successful response or one of the following fault codes in Table 8.1.6.1-1.

Table 8.1.6.1-1: SetParameterAttributes Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)

8.1.6.2 Notification primitive mapping

Notify Request and Response primitives shall map to the BBF TR-069 [4] notification mechanism. CPEs produce notifications for subscribed attributes using the BBF TR-069 [4] Inform method, the Inform method has an argument Event that has as one of the EventCodes with the value "4 VALUE CHANGE" indicating that a subscribed parameter's value has changed. The parameter(s) that have changed are included ParameterList argument of the Inform method.

The ParameterList argument is list of name-value pairs; the name is parameter name and shall be mapped to the objectPath attribute of the Resource while the value is the most recent value of the parameter.

NOTE: BBF TR-069 [4] CPEs do not report value changes of parameters that were modified by the ACS.

8.2 <mgmtCmd> and <execInstance> resource primitive mappings

8.2.1 Update (Execute) primitive for the <mgmtCmd> resource

8.2.1.0 Introduction

When the Update Request primitive for <mgmtCmd> resource addresses the execEnable attribute of the <mgmtCmd> resource, it effectively triggers an Execute <mgmtCmd> procedure.

The Hosting CSE performs command conversion of its <execInstance> sub-resources. The mapping between the <execInstance> attributes and the BBF TR-069 [4] RPC procedures triggered is based on the value of the cmdType attribute of the <mgmtCmd> resource defined in Table 8.2.1.0-1. The CPE acceptance of the corresponding RPC procedures is indicated by returning a successful Response primitive to the initial Update Request.

The Fault Codes which may be returned by the CPE to the Hosting CSE are mapped onto execResult codes and stored in the corresponding <execInstance> attributes, and are detailed in clauses 8.2.1.1 to 8.2.1.7 inclusive.

Table 8.2.1.0-1 Mapping of Execute <mgmtCmd> primitives to BBF TR-069 [4] RPC

cmdType value	BBF TR-069 RPCs
"DOWNLOAD"	Download RPC (see clause 8.2.1.1) and TransferComplete RPC (clause 8.2.1.3)
"UPLOAD"	Upload RPC (clause 8.2.1.2) and T ransferComplete RPC (clause 8.2.1.3)
"SOFTWAREINSTALL"	ChangeDUState RPC (clause 8.2.1.4) and ChangeDUStateComplete RPC (clause 8.2.1.5)
"SOFTWAREUNINSTALL"	ChangeDUState RPC (clause 8.2.1.4) and ChangeDUStateComplete RPC (clause 8.2.1.5)
"REBOOT"	Reboot RPC (clause 8.2.1.6)
"RESET"	Factory reset RPC (clause 8.2.1.7)

8.2.1.1 Execute File Download

The download file transfer operation may use the Download mechanism defined in BBF TR-069 [4]. The Download mechanism is an asynchronous command which returns a successful response or one of the following fault codes mapped onto execResult values as detailed in Table 8.2.1.1-1. A successful response to the Update primitive triggering the Execute procedure means that the CPE has accepted the Download RPC.

Table 8.2.1.1-1: Download Fault Code Mapping

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST_DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9003	Invalid arguments	STATUS_INVALID_ARGUMENTS
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error).	STATUS_RESOURCES_EXCEEDED
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	STATUS_FILE_TRANSFER_FAILED
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods, not associated with Scheduled Download method).	STATUS_FILE_TRANSFER_SERVER_AUTHENTICATION_FAILURE
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_UNSUPPORTED_PROTOCOL

8.2.1.2 Execute File Upload Operations

The upload file transfer operation shall use the Upload mechanism defined in BBF TR-069 [4]. The Upload mechanism is an asynchronous command that consists of the synchronous Upload RPC for the Upload and the asynchronous TransferComplete RPC. The Upload RPC returns a successful response or one of the following fault codes mapped onto execResult values as detailed in Table 8.2.1.2-1. A successful response to the Update primitive triggering the execute procedure means that the CPE has accepted the Upload RPC in Table 8.2.1.2-1.

Table 8.2.1.2-1: Upload Fault Code Mapping

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9003	Invalid arguments	STATUS_INVALID_ARGUMENTS
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	STATUS_RESOURCES_EXCEEDED
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	STATUS_UPLOAD_FAILED
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_SERVER_AUTHENTICATION_FAILURE
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_UNSUPPORTED_PROTOCOL

8.2.1.3 Report Results using TransferComplete RPC

After a File Download or Upload has been attempted, the result of the operation is reported using the TransferComplete RPC. The TransferComplete RPC indicates a successful operation or one of the following fault codes mapped onto execResult values in Table 8.2.1.3-2.

Table 8.2.1.3-2: TransferComplete Fault Code Mapping

Fault code	Description	execResult Code
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	STATUS_FILE_TRANSFER_FAILED
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	STATUS_UPLOAD_FAILED
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_SERVER_AUTHENTICATION_FAILURE
9014	File transfer failure: unable to join multicast group (associated with Download, TransferComplete or AutonomousTransferComplete methods).	STATUS_FILE_TRANSFER_FAILED_MULTICAST_GROUP_UNABLE_JOIN
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_SERVER_CONTACT_FAILED
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_FILE_ACCESS_FAILED
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_DOWNLOAD_INCOMPLETE
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_FILE_CORRUPTED
9019	File transfer failure: file authentication failure (associated with Download, TransferComplete or AutonomousTransferComplete methods).	STATUS_FILE_TRANSFER_FILE_AUTHENTICATION_FAILURE
9020	File transfer failure: unable to complete download within specified time windows (associated with TransferComplete method).	STATUS_FILE_TRANSFER_WINDOW_EXCEEDED

8.2.1.4 Execute Software Operations with ChangeDUState RPC

The software installation and uninstall operations shall use the ChangeDUState mechanism defined in BBF TR-069 [4]. The ChangeDUState mechanism is an asynchronous command that consists of the synchronous ChangeDUState RPC and returns a successful response or one of the fault codes mapped onto execResult values as detailed in Table 8.2.1.4-1. A successful response to the Update primitive triggering the Execute procedure means that the CPE has accepted the ChangeDUState RPC.

Table 8.2.1.4-1: ChangeDUState Fault Code Mapping

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	STATUS_RESOURCES_EXCEEDED

8.2.1.5 Report Results with ChangeDUStateComplete RPC

After software installation and uninstall operations using a ChangeDUState mechanism as defined in BBF TR-069 [4], the result of the state change operation is retrieved using the ChangeDUStateComplete RPC. The ChangeDUStateComplete RPC indicates a successful operation or one of the fault codes mapped onto execResult values as detailed in Table 8.2.1.5-1.

Table 8.2.1.5-1: ChangeDUStateComplete Fault Code Mapping

Fault code	Description	execResult Code
9001	Request denied (no reason specified)	STATUS_REQUEST_DENIED
9003	Invalid arguments	STATUS_INVALID_ARGUMENTS
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_SERVER_AUTHENTICATION_FAILURE
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_UNSUPPORTED_PROTOCOL
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_SERVER_CONTACT_FAILED
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_FILE_ACCESS_FAILED
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_DOWNLOAD_INCOMPLETE
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	STATUS_FILE_TRANSFER_FAILED_FILE_CORRUPTED
9022	Invalid UUID Format (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	STATUS_INVALID_UUID_FORMAT
9023	Unknown Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	STATUS_UNKNOWN_EXECUTION_ENVIRONMENT
9024	Disabled Execution Environment (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update, and Uninstall)	STATUS_DISABLED_EXECUTION_ENVIRONMENT
9025	Deployment Unit to Execution Environment Mismatch (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	STATUS_EXECUTION_ENVIRONMENT_MISMATCH
9026	Duplicate Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install only)	STATUS_DUPLICATE_DEPLOYMENT_UNIT
9027	System Resources Exceeded (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install and Update)	STATUS_SYSTEM_RESOURCES_EXCEEDED

Fault code	Description	execResult Code
9028	Unknown Deployment Unit (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update and Uninstall)	STATUS_UNKNOWN_DEPLOYMENT_UNIT
9029	Invalid Deployment Unit State (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Install, Update and Uninstall)	STATUS_INVALID_DEPLOYMENT_UNIT_STATE
9030	Invalid Deployment Unit Update - Downgrade not permitted (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	STATUS_INVALID_DEPLOYMENT_UNIT_UPDATE_DOWNGRADE_DISALLOWED
9031	Invalid Deployment Unit Update - Version not specified (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	STATUS_INVALID_DEPLOYMENT_UNIT_UPDATE_UPGRADE_DISALLOWED
9032	Invalid Deployment Unit Update - Version already exists (associated with DUStateChangeComplete or AutonomousDUStateChangeComplete methods: Update only)	STATUS_INVALID_DEPLOYMENT_UNIT_UPDATE_VERSION_EXISTS

8.2.1.6 Execute Reboot operation

The reboot operation shall use the Reboot RPC defined in BBF TR-069 [4]. The Reboot RPC is a synchronous command. A successful response to the Update primitive triggering the Execute procedure means that the CPE has accepted the Reboot RPC. The Reboot RPC returns a successful response or one of the fault codes mapped onto execResult values as detailed in Table 8.2.1.6-1.

Table 8.2.1.6-1: Reboot Fault Code Mapping

Fault code	Description	execResult Code
9001	Request denied (no reason specified)	STATUS_REQUEST_DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9003	Invalid arguments	STATUS_INVALID_ARGUMENTS

8.2.1.7 Execute Factory Reset operation

The factory reset operation shall use the FactoryReset RPC defined in BBF TR-069 [4]. The FactoryReset RPC is a synchronous command. A successful response to the Update primitive triggering the Execute procedure means that the CPE has accepted the FactoryReset RPC. The FactoryReset RPC returns a successful response or one of the fault codes mapped onto execResult values as detailed in Table 8.2.1.7-1.

Table 8.2.1.7-1: FactoryReset Fault Code Mapping

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST_DENIED
9002	Internal error	STATUS_INTERNAL_ERROR
9003	Invalid arguments	STATUS_INVALID_ARGUMENTS

8.2.2 Delete <mgmtCmd> resource primitive mapping

The Delete Request primitive for the <mgmtCmd> resource may initiate BBF TR-069 [4] RPC commands for the corresponding <execInstance> sub-resources as follows:

- If there are no <execInstance> sub-resources with RUNNING execStatus, a successful response to the Delete primitive is returned and the <mgmtCmd> resource is deleted without triggering any BBF TR-069 [4] RPCs.
- If there are <execInstance> sub-resources with RUNNING execStatus that resulted in cancellable BBF TR-069 [4] RPCs (e.g. File Upload and File Download RPCs), a BBF TR-069 [4] CancelTransfer RPC shall be initiated for each cancellable operation. Upon completion of all the cancellation operations, if any fault codes are returned by the CPE, an unsuccessful Response to the Delete primitive with status code "Delete mgmtCmd- execInstance cancellation error" is returned, and the <mgmtCmd> resource is not deleted. The execStatus attribute of each specific <execInstance> is set to CANCELLED and the execResult attribute is set to "STATUS_SUCCESS" for successful RPCs. For the unsuccessful case, execResult is determined from the RPC fault codes as detailed in Table 8.2.2-1. If all cancellation operations are successful on the managed entity, a successful Response to the Delete primitive is returned and the <mgmtCmd> resource is deleted.
- If there is at least one <execInstance> sub-resource with RUNNING execStatus that resulted in non-cancellable BBF TR-069 [4] RPCs (e.g. RPCs other than File Upload and File Download RPCs), the execStatus attribute of the specific <execInstance> is changed to STATUS_NON_CANCELLABLE. An unsuccessful Response to the Delete primitive with status code "Delete mgmtCmd- execInstance cancellation error" is returned and the <mgmtCmd> resource is not deleted.

Table 8.2.2-1: CancelTransfer Fault Code Mapping for Delete <mgmtCmd>

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9021	Cancellation of file transfer not permitted in current transfer state	STATUS_CANCELLATION_DENIED

8.2.3 Update (Cancel) <execInstance> primitive mapping

When the Update Request primitive for an <execInstance> sub-resource addresses the execDisable attribute of the <execInstance > sub-resource, it effectively triggers a Cancel <execInstance> resource procedure.

The hosting CSE determines whether the <execInstance> resource has a RUNNING execStatus and weather the resulting BBF TR-069 [4] RPCs are cancellable. Currently, only the BBF TR-069 File Upload and File Download RPCs are cancellable using the BBF TR-069 [4] CancelTransfer RPC:

- If the addressed <execInstance> sub-resource has an execStatus other than RUNNING, an un-successful Response to the Update primitive is returned with status code "Cancel execInstance - already complete".
- If the addressed <execInstance> sub-resources has RUNNING execStatus and resulted in cancellable BBF TR-069 [4] RPCs (e.g. File Upload and File Download RPCs), a BBF TR-069 [4] CancelTransfer RPC shall be initiated. For a successful CancelTransfer RPC the execStatus attribute of the specific <execInstance> is set to CANCELLED and a successful Response is sent to the Update primitive. For a successful CancelTransfer RPC the execStatus attribute of the specific <execInstance> is set to CANCELLED, the execResult attribute is set to "STATUS_SUCCESS" and a successful Response is sent to the Update primitive. For an unsuccessful CancelTransfer RPC the execResult attribute is determined from the RPC fault codes as detailed in Table 8.2.3-1 and an unsuccessful Response is sent to the Update primitive with status code "Cancel execInstance - cancellation error".
- If the addressed <execInstance> sub-resources has RUNNING execStatus and resulted non-cancellable BBF TR-069 [4] RPCs (e.g. RPCs other than File Upload and File Download RPCs), the execStatus attribute of the specific <execInstance> is changed to STATUS_NON_CANCELLABLE. An unsuccessful Response is sent to the Update primitive with status code "Cancel execInstance - not cancellable".

Table 8.2.3-1: CancelTransfer Fault Code Mapping for Update (Cancel) <execInstance>

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9021	Cancellation of file transfer not permitted in current transfer state	STATUS_CANCELLATION_DENIED

8.2.4 Delete <execInstance> primitive mapping

The Delete Request primitive for an <execInstance> sub-resource may initiate BBF TR-069 [4] RPC commands for the corresponding <execInstance> sub-resources as follows:

- If the addressed <execInstance> sub-resource has an execStatus other than RUNNING, an successful Response to the Delete primitive is returned and the <execInstance> sub-resource is deleted without triggering any BBF TR-069 [4] RPCs.
- If the addressed <execInstance> sub-resource has RUNNING execStatus and resulted in cancellable BBF TR-069 [4] RPCs (e.g. File Upload and File Download RPCs), a BBF TR-069 [4] CancelTransfer RPC shall be initiated. For a successful CancelTransfer RPC a successful response is sent to the Delete primitive and the <execInstance> sub-resource is deleted. For an unsuccessful CancelTransfer RPC the execStatus attribute is determined from the RPC fault codes as detailed in Table 8.2.4-1 and an unsuccessful Response is sent to the Delete primitive with status code "Delete execInstance - cancellation failed".
- If the addressed <execInstance> sub-resource has RUNNING execStatus and resulted non-cancellable BBF TR-069 [4] RPCs (e.g. RPCs other than File Upload and File Download RPCs), the execResult attribute is set to STATUS_NON_CANCELLABLE and an unsuccessful Response is sent to the Update primitive with status code "Delete execInstance - not cancellable".

Table 8.2.4-1: CancelTransfer Fault Code Mapping for Delete <execInstance>

Fault code	Description	execResult Code
9000	Method not supported	STATUS_REQUEST_UNSUPPORTED
9001	Request denied (no reason specified)	STATUS_REQUEST DENIED
9021	Cancellation of file transfer not permitted in current transfer state	STATUS_CANCELLATION_DENIED

8.3 Resource [myCertFileCred] primitive mappings

8.3.1 Introduction

This clause contains information regarding the procedures for establishing a certificates presented by the managed entity in order for the peer to authenticate the managed entity.

8.3.2 Creation of Resource [myCertFileCred]

8.3.2.1 Introduction

The creation of a [myCertFileCred] resource requires the use of the BBF TR-069 [4] Download RPC to establish the credential on the managed entity. Once the managed entity has obtained the credential, the Device.Security.Certificate.{i} instance's SUIDs parameter is set from the [myCertFileCred] attribute using the BBF TR-069 [4] Set RPC.

8.3.2.2 Procedure for creation of Resource [myCertFileCred]

The Create Request and Response primitives for Resource [myCertFileCred] that results in a download file transfer shall use the Download mechanism defined in BBF TR-069 [4]. The Download mechanism is an asynchronous command that consists of the synchronous Download RPC for the Download and the asynchronous TransferComplete RPC. The Download RPC returns a successful response or one of the following fault codes in Table 8.3.2.2-1. A successful response means that the CPE has accepted the Download RPC.

Table 8.3.2.2-1: Download Fault Code Mapping

Fault code	Description	Response Status Code
9000	Method not supported	4000 (BAD_REQUEST)
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9003	Invalid arguments	4000 (BAD_REQUEST)
9004	Resources exceeded (when used in association with SetParameterValues, this cannot be used to indicate Parameters in error)	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9013	Unsupported protocol for file transfer (associated with Upload, Download, ScheduleDownload, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)

Once the CPE has attempted to download the file, the CPE reports the result of the download operation using the TransferComplete RPC. The TransferComplete RPC indicates a successful operation or one of the following fault codes in Table 8.3.2.2-2.

Table 8.3.2.2-2: TransferComplete Fault Code Mapping

Fault code	Description	Response Status Code
9001	Request denied (no reason specified)	4000 (BAD_REQUEST)
9002	Internal error	4000 (BAD_REQUEST)
9010	File transfer failure (associated with Download, ScheduleDownload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9011	Upload failure (associated with Upload, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9012	File transfer server authentication failure (associated with Upload, Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9014	File transfer failure: unable to join multicast group (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9015	File transfer failure: unable to contact file server (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9016	File transfer failure: unable to access file (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9017	File transfer failure: unable to complete download (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9018	File transfer failure: file corrupted or otherwise unusable (associated with Download, TransferComplete, AutonomousTransferComplete, DUStateChangeComplete, or AutonomousDUStateChangeComplete methods).	4000 (BAD_REQUEST)
9019	File transfer failure: file authentication failure (associated with Download, TransferComplete or AutonomousTransferComplete methods).	4000 (BAD_REQUEST)
9020	File transfer failure: unable to complete download within specified time windows (associated with TransferComplete method).	4000 (BAD_REQUEST)

Upon successful TransferComplete notification from the CPE, the newly created Device.Security.Certificate.{i} object instance shall be assigned the values of the SUIDs attribute using the procedure for updating parameters in clause 8.1.4.

9 Server Interactions

9.0 Introduction

This clause specifies how the IN-CSE interacts with an ACS in order to manage the Resources described in the present document. The IN-CSE interaction with an ACS includes:

- Establishment of the communication session between the IN-CSE and ACS.
- Processing of requests and notifications between the IN-CSE and the ACS.
- Discovery.

NOTE: The Broadband Forum has not defined a protocol specification for the Northbound Interface of an ACS. As such, the present document only describes the expectations of this interface in the form of requirements on the ACS.

9.1 Communication Session Establishment

9.1.1 IN-CSE to ACS Communication Session Establishment

When the IN-CSE detects that it has to delegate an interaction with a device resource to an ACS, the IN-CSE establishes a communication session with the ACS. The establishment of a communication session between the IN-CSE and ACS provides security dimensions for Access control, Authentication, Non-repudiation, Data confidentiality, Communication security, Data integrity and Privacy adhering to the BBF TR-131 [7] Architectural requirement A7.

The IN-CSE may establish multiple sessions with an ACS based on the security model utilized between the IN-CSE and the ACS.

9.1.2 ACS to IN-CSE Communication Session Establishment

When the ACS detects a change to resources it manages that the IN-CSE has expressed interest, the ACS requests the IN-CSE to establish a session if a session does not exist for the resource being managed. The establishment of a communication session between the IN-CSE and ACS provides security dimensions for Access control, Authentication, Non-repudiation, Data confidentiality, Communication security, Data integrity and Privacy adhering to the BBF TR-131 [7] Architectural requirement A7.

The ACS may establish multiple sessions with an IN-CSE based on the security model utilized between the IN-CSE and the ACS.

While a session between the ACS and IN-CSE is not established, the ACS retains any notifications or changes in the resources based on an Event retention policy (i.e. time, number of events).

When an ACS to IN-CSE interaction is required and a session does not exist, the ACS requests to initiate a session based on a Session Initiation Policy (i.e. Periodic contact establishment (schedule), upon event detection with timeframe window).

9.1.3 ACS and IN-CSE Communication Session Requirements

When establishing a session from the ACS to the IN-CSE:

- If a session does not exist between the IN-CSE and ACS, the ACS shall retain any notifications or changes in the resources based on an Event retention policy (i.e. time, number of events).

- When an ACS to IN-CSE interaction is required and a session does not exist, the ACS shall be capable to initiate a session based on a Session Initiation Policy (i.e. Periodic contact establishment (schedule), upon event detection with timeframe window).

9.2 Processing of Requests and Responses

9.2.1 Request and Notification Formatting

Requests and Notifications mechanisms between the IN-CSE and the DM Server format the XML schema of the CPE methods defined in BBF TR-069 [4] as an ACS would format the CPE methods that it would pass to the CPE. The IN-CSE would then also process the CPE methods as defined in BBF TR-069 [4]. Likewise the ACS would send notifications in the format of the XML schema of the CPE for sending events using the Inform RPC.

9.2.2 ACS Request Processing Requirements

When receiving requests from the IN-CSE the ACS shall be capable of defining mechanisms to support triggering of immediate operations to device. If the device is not available the ACS returns an appropriate error code.

The ACS shall provide capability for the IN-CSE to indicate request policies to include: Retry policy, Request Time out.

9.2.3 ACS Notification Processing Requirements

When sending notifications to the IN-CSE:

- The ACS shall be capable of providing a mechanism for the IN-CSE to subscribe to events.
- The ACS shall be capable of providing a list of events for which the IN-CSE can subscribe.
- The ACS shall be capable of providing a mechanism for the IN-CSE to unsubscribe from events.
- The ACS shall be capable of providing an event delivery mechanism.
- The ACS shall be capable of providing the capability for the IN-CSE to request event filters including: Event Code; Specific parameters changing value; Device; Any combination of the previous criteria.
- The IN-CSE shall be capable of subscribing to be notified of changes to resources it manages.
- The ACS shall be capable of notifying the IN-CSE of changes to resources to which the client has subscribed.

9.3 Discovery and Synchronization of Resources

For devices under management, the IN-CSE may discover resources of interest (metadata and values) within a device using the ACS.

For resources of interest, the IN-CSE may also express an interest to be notified of a resource if a resource is changed (added, deleted, updated).

The IN-CSE shall be capable to discover and subscribe to changes of resources in order to synchronize the IN-CSE with resources of interest of the ACS.

9.4 Access Management

9.4.0 Introduction

Once a request has performed an Access Decision by the IN-CSE to allow the request, the IN-CSE shall select the appropriate ACS along with elements the ACS would need to implement access management within the ACS. These would include the Identity of the subject (oneM2M Originator) of the request which is needed in scenarios where the original issuer of the request is needed to be known - this could be done by correlating principals (e.g. Roles, Accounts) used by the IN-CSE and ACS.

9.4.1 Access Management Requirements

- The ACS shall be capable of providing a mechanism for the IN-CSE to discover the Access Management elements used to authorize and authenticate access to resources controlled by the ACS.
- The IN-CSE shall be capable of correlating Access Management elements provided by the ACS to Access Management elements used by the IN-CSE.
- The IN-CSE shall be capable of providing secured storage of Access Management elements within the IN-CSE.

10 New Management Technology Specific Resources

BBF TR-181 [6] provides a list of management objects that have been standardized by the Broadband Forum and where possible, clause 7 provides a mapping of the Resources to standardized management objects. The oneM2M vendor specific extensions to the BBF TR-181 [6] data model are provided in the ts-0006-3-3-0.xml available from https://git.onem2m.org/MAS/TR69Model.

History

Version	Date	Status
V5.0.1	August 2026	Publication